SPACE X · WORD BANK

Word bank, concept level — S1–S3 (v4)

32 core terms. Each one answers five questions, in this order, every time.

How to read an entry (and how to write it in your notebook)

- **Name** in bold first, then **Also called**. Say the official name out loud before anything else; that's the rule.
- **Meaning** = the one-line definition you'd put in a dictionary. This is what you've been copying so far. It is not enough on its own.
- **Concept** = *why the thing exists*: what problem it solves, what would go wrong without it. This is the part that makes the word stick, because now it has a job.
- **Numbers** = one real example with a real value. A term with no number attached is a term you can't check yourself.
- **Not this** = the thing people mix it up with, and the one-line test that separates them.
- **Connects** = which formulas or other terms it plugs into. This is how you walk from one idea to the next in a conversation instead of stalling.

In the notebook: the back word-bank pages only have room for Name · Also called · Meaning (one line). Put Concept and Numbers in the margin of the lecture page the term came from, or say them out loud when you review. Don't copy all five lines for all 32 terms; the app has them.

Order of the entries: by scene, in the order the app shows them (1 Falling & missing → 7 The free lunch). The number badge is the scene.

Test yourself: cover everything but the name. Say the concept line. If you can't say *why it exists*, you don't own the word yet, even if you can define it.

WORD BANK · 1/8

Concept-level word bank

Centripetal force

1

Also called: center-seeking force

MEANING The inward force that bends straight-line motion into a curve. For a satellite, gravity is the centripetal force; for a ball on a string, the string's tension. Without it, motion continues in a straight line (Newton's first law).

CONCEPT Nothing moves in a circle on its own; something must pull it inward constantly, or it goes straight. For a satellite that something is gravity. Seeing gravity as the string, not the enemy, is the key to understanding why orbits need no engine.

NUMBERS ISS: gravity provides 8.7 m/s² inward; at 7.67 km/s that's exactly the v^2/r a 6,771 km circle requires.

NOT THIS 'Centrifugal force,' the outward feeling, which isn't a real force; it's your inertia resisting the inward pull.

CONNECTS Circular velocity formula comes from setting gravity equal to the required centripetal pull.

Station-keeping

1

Also called: orbit maintenance

MEANING Small periodic burns that cancel perturbations (drag in LEO, the Moon's and Sun's gravity in GEO) so the orbit stays where the mission needs it. Sets a GEO satellite's useful life. Not what 'holds the satellite up'; nothing does.

CONCEPT Orbits aren't perfectly stable: drag, the Moon, the Sun and Earth's lumpy shape all nudge a satellite off its intended path. Station-keeping is the routine of small burns that cancel those nudges. It's maintenance, and it's what actually ends most satellites' lives, because it uses fuel.

NUMBERS GEO: ~50 m/s per year, burns every few weeks. ISS: a reboost of a few m/s roughly monthly.

NOT THIS 'Holding the satellite up.' Nothing holds it up; station-keeping only corrects drift from where it would be anyway.

CONNECTS Reboost, perturbations, graveyard orbit when fuel runs out.

Geostationary orbit (GEO)

2

Also called: Clarke orbit

MEANING The special geosynchronous orbit that is also circular and equatorial (i = 0), at 35,786 km altitude (42,164 km radius), so the satellite appears fixed. TV and weather satellites.

CONCEPT A satellite that doesn't move in the sky lets you point a fixed dish at it forever. That needs a lap time of exactly one Earth rotation, which happens at only one altitude, and it needs the orbit to lie over the equator, because the equator is the only ground circle centered on Earth's center.

NUMBERS Altitude 35,786 km, radius 42,164 km, speed 3.07 km/s, period 23 h 56 m. Three satellites cover almost all of Earth except the poles.

NOT THIS Geosynchronous: same period but tilted, so it traces a figure-8 instead of staying put.

CONNECTS Kepler's third law gives the altitude; station-keeping holds the slot; GTO gets you there.

Constellation

2

MEANING A set of satellites in coordinated orbits that function as one system (GPS, Iridium, Starlink). Arranged in several orbital planes so coverage is continuous.

CONCEPT One satellite can only be in one place, and low ones move fast. For continuous coverage of the whole planet you need enough satellites, spread over enough orbital planes, that one is always overhead wherever you are. That coordinated swarm is a constellation.

NUMBERS GPS 24+ in 6 planes. Iridium 66 in 6 polar planes. Starlink thousands in dozens of planes.

NOT THIS A group of satellites that merely share an orbit. A constellation is designed as one system with coverage guarantees.

CONNECTS Footprint per satellite, revisit time, orbital planes (Walker).

Concept-level word bank

Classical Orbital Elements (COEs) 3

Also called: Keplerian elements · orbital parameters · the six elements

MEANING The six numbers that completely describe an orbit: a, e, i, Ω, ω, ν. Five fix the shape and orientation; the sixth says where the body is. The hula hoop's six settings.

CONCEPT Position and velocity (six numbers) describe where a satellite is right now, but they change every second. Rewrite the same information as six numbers that describe the orbit itself, and five of them barely change for years. That's the whole point: a stable description of a path instead of a snapshot of a point.

NUMBERS ISS: a ≈ 6,790 km, e ≈ 0.0005 (nearly circular), i = 51.6°, Ω and ω drift slowly, ν changes 360° every 92 minutes.

NOT THIS Not the same as latitude/longitude, which describe a point on the ground. Elements describe the whole path in space.

CONNECTS Vis-viva uses a; period uses a; the burn rules say which element each burn moves.

Semi-major axis (a) 3

Also called: half the major axis · 'size' of the orbit

MEANING Half the longest width of the ellipse; the average of perigee and apogee distances from the center of the body. Sets the orbit's size and, through Kepler's third law, its period.

CONCEPT One number that captures how big the orbit is. Because period depends only on a (Kepler's third law), a tells you how long a lap takes without knowing anything else.

NUMBERS ISS a = 6,790 km → 92 min. GPS a = 26,560 km → 12 h. GEO a = 42,164 km → 24 h.

NOT THIS Altitude. a is measured from Earth's center and is the average of the near and far distances; altitude is height above the ground at one point.

CONNECTS Kepler's third law, vis-viva, Hohmann transfer (a of the transfer ellipse = $(r_1 + r_2)/2$).

Eccentricity (e) 3

Also called: 'shape' of the orbit

MEANING How far the ellipse departs from a circle. 0 = circle, 0 < e < 1 = ellipse, e = 1 = parabola (escape), e > 1 = hyperbola (flyby).

CONCEPT Orbits are ellipses, and an ellipse needs a 'how squashed' number. e is that number, built so 0 is a circle and 1 is the edge of not closing at all. Everything you need to know about shape is in one dimensionless number.

NUMBERS ISS 0.0005 (circle). Molniya 0.74 (cigar). GTO 0.73. New Horizons leaving Earth: > 1 (hyperbola, never comes back).

NOT THIS Inclination. e is squash, i is tilt. An orbit can be very squashed and flat, or circular and steeply tilted.

CONNECTS Perigee/apogee distances: r_p = a(1−e), r_a = a(1+e).

Inclination (i) 3

Also called: tilt

MEANING Angle between the orbital plane and the reference plane (Earth's equator for Earth satellites). 0° equatorial, 90° polar, > 90° retrograde.

CONCEPT Where on Earth a satellite can fly over is set entirely by tilt: a satellite never goes to a latitude higher than its inclination. So i is chosen by the mission (what must you see?) and is the most expensive element to change afterward.

NUMBERS GEO 0° (equator only). ISS 51.6° (sees up to 51.6° N/S). Polar/sun-synchronous ~98° (sees everything).

NOT THIS Ω. Both are angles describing orientation; i is how much the plane is tilted, Ω is which way the tilt faces.

CONNECTS Plane change Δv = 2v·sin(Δi/2); launch latitude sets the minimum i you can reach cheaply.

Concept-level word bank

Right ascension of the ascending node (Ω, RAAN) 3

Also called: longitude of the ascending node · 'swivel'

MEANING Angle, measured in the reference plane from the vernal equinox (♈), to the point where the orbit crosses the plane heading north.

CONCEPT A tilted plane can face any direction around Earth's axis. Ω pins that direction down by saying where the orbit crosses the equator going north, measured from a fixed star-based reference. Without it two orbits with the same tilt could be at right angles to each other.

NUMBERS GPS uses six planes with Ω spaced 60° apart so satellites are spread around the globe rather than bunched.

NOT THIS Argument of periapsis ω. Ω is where the plane crosses the equator (measured in the equator plane); ω is where perigee sits (measured inside the orbit plane).

CONNECTS J_2 makes Ω drift; sun-synchronous orbits exploit that drift.

Argument of periapsis (ω) 3

Also called: argument of perigee · 'spin in plane'

MEANING Angle, measured in the orbital plane, from the ascending node to periapsis. Says which way the long axis of the ellipse points.

CONCEPT An elliptical orbit has a near end and a far end, and it matters where the near end points: over the target you want to photograph up close, or over the region you want to linger above. ω records that.

NUMBERS Molniya orbits set ω ≈ 270° so apogee hangs over the northern hemisphere for hours.

NOT THIS Ω (see above). Also 'perigee' itself: ω is the angle to perigee, not the perigee distance.

CONNECTS Radial burns rotate ω; J_2 makes ω drift for most inclinations.

True anomaly (ν or θ) 3

MEANING Angle from periapsis to the body's current position, measured at the central body. The only element that changes quickly as the orbit proceeds.

CONCEPT Five elements freeze the orbit; one number is still needed to say where on it the satellite is at this instant. That's ν. It's the only element that changes fast, which is why the other five are the useful 'address' and ν is the 'time.'

NUMBERS ν = 0° at perigee, 180° at apogee. The ISS sweeps all 360° every 92 minutes.

NOT THIS Mean anomaly M, which grows evenly with time; ν does not, because the satellite is faster near perigee.

CONNECTS Kepler's second law explains why ν changes unevenly.

Periapsis / apoapsis 3

Also called: perigee/apogee (Earth) · perihelion/aphelion (Sun) · perilune/apolune (Moon) · periareion/apoareion (Mars)

MEANING The closest and farthest points of an orbit from the central body. Fastest at periapsis, slowest at apoapsis.

CONCEPT Any non-circular orbit has one nearest and one farthest point, and they're where the physics is simplest: speed is purely sideways there, which is why burns are planned there and why their distances define a and e.

NUMBERS GTO: perigee 300 km altitude at 10.15 km/s, apogee 35,786 km at 1.61 km/s.

NOT THIS The suffix changes with the body: -gee Earth, -helion Sun, -lune Moon, -apsis generic. Same two points.

CONNECTS Burn rules: prograde at periapsis raises apoapsis; at apoapsis raises periapsis.

WORD BANK · 4/8

Concept-level word bank

Standard gravitational parameter (μ)

3

Also called: GM · 'mu'

MEANING The body's mass times the gravitational constant, in one number: Earth μ = 398,600 km³/s². Shows up in every orbit formula.

CONCEPT Every orbit formula has G × M in it, and G × M is known far more precisely than either G or M alone. So engineers bundle it into one constant per body and never separate it.

NUMBERS Earth 398,600 km³/s². Sun 1.327 × 10^{11}. Mars 42,828. Jupiter 1.267 × 10^{8}.

NOT THIS g (9.8 m/s²), which is surface gravity, a different thing. μ/r² gives g at any distance.

CONNECTS Appears in vis-viva, period, escape speed, turning angle.

Vis-viva equation

3

Also called: orbital energy equation

MEANING $v^2 = \mu(2/r - 1/a)$. Speed anywhere on any orbit from distance and semi-major axis. Gives circular speed (a = r) and escape speed (a → ∞).

CONCEPT The energy of an orbit is fixed, so speed and distance trade off along it. Vis-viva is that trade-off written as one formula: know how far you are and how big the orbit is, and you know how fast you're going. It replaces a dozen special cases with one line.

NUMBERS GTO at perigee: $v^2 = 398{,}600 \cdot (2/6{,}671 - 1/24{,}418) \rightarrow v = 10.15$ km/s. At apogee: 1.61 km/s.

NOT THIS Circular velocity $v = \sqrt{(\mu/r)}$ is just vis-viva with a = r. Escape velocity is vis-viva with a = ∞.

CONNECTS Every Δv calculation: speed before and after a burn come from vis-viva.

Stefan–Boltzmann law

4

MEANING Radiated power per area = $\sigma \cdot \varepsilon \cdot T^4$. Doubling absolute temperature radiates 16× more. The physics behind radiator sizing.

CONCEPT Radiation is the only way heat leaves a spacecraft, and this law says how fast: proportional to the fourth power of absolute temperature. The steep T^4 means radiators work far better when hot, cold things barely radiate, and sizing a radiator is a one-line calculation.

NUMBERS 1 m² at 300 K, ε = 0.9: ~410 W. At 350 K: ~760 W. At 200 K: only ~80 W.

NOT THIS Conduction/convection, which depend on a temperature difference, not an absolute temperature to the fourth.

CONNECTS Radiator sizing, emissivity/absorptivity coatings, why hot spots persist.

Outgassing

4

MEANING Release of trapped gas and volatiles from materials in vacuum. Measured as TML (total mass loss) and CVCM (collected volatile condensable material); spacecraft materials must stay under 1% and 0.1%.

CONCEPT Every material made in air has gas dissolved and trapped inside it. In vacuum there's nothing holding it in, so it slowly leaks out, and in vacuum vapor travels in straight lines to the coldest surface it can find: your lens. So materials are chosen and baked for low outgassing before launch.

NUMBERS Limits: TML < 1%, CVCM < 0.1% (ASTM E595, 125 °C for 24 h). Bake-outs run days at 100+ °C.

NOT THIS Leaks from tanks or cabins. Outgassing is from solid materials themselves: plastics, adhesives, paints.

CONNECTS Contamination of optics, clean rooms, MLI material choice.

WORD BANK · 5/8

Concept-level word bank

Spacecraft charging

4

Also called: surface charging · electrostatic discharge (ESD)

MEANING Build-up of electric charge on surfaces with no path to bleed it off, until it arcs. Mitigated by conductive coatings and bonding (grounding straps).

CONCEPT On Earth static bleeds off into air and ground. In space it can't, so charge collects: sunlight knocks electrons off lit surfaces (positive) while plasma electrons land on shaded ones (negative). Two surfaces can end up kilovolts apart and arc, which fries electronics. Every surface is therefore made conductive and strapped to every other.

NUMBERS Differential potentials of several kV have been measured on GEO satellites during geomagnetic storms; arcs have killed solar-array strings.

NOT THIS Radiation damage. Charging is static electricity; radiation is energetic particles passing through chips.

CONNECTS Photoelectric effect, ESD, conductive coatings (ITO), bonding straps.

Disturbance torque

4

Also called: perturbing torque

MEANING A small twisting force on a spacecraft: gravity-gradient, solar radiation pressure, aerodynamic, magnetic.

CONCEPT Even in 'empty' space tiny twists never stop: gravity pulls the near end harder, sunlight pushes on big panels, leftover air drags unevenly, Earth's magnetism tugs. Each is tiny, none ever stops, nothing damps them. Attitude control exists because of them.

NUMBERS Small satellite: ~10^{-5} N·m. ISS: several N·m, enough to tumble it within hours if uncontrolled.

NOT THIS Orbital perturbations, which change the path. Disturbance torques change the pointing.

CONNECTS Reaction wheels counter them; momentum dumping resets the wheels.

Reaction wheel / momentum wheel / control moment gyro (CMG)

4

MEANING Internal spinning masses used to turn or stabilize a spacecraft without fuel; CMGs tilt a spinning wheel for much larger torque.

CONCEPT In vacuum there's nothing to push against to turn, so you push against yourself: spin a wheel inside and the spacecraft turns the other way. No fuel, very precise. The catch is that constant one-way pushes make the wheels spin ever faster until they saturate.

NUMBERS Hubble uses four reaction wheels for pointing to 0.007 arcseconds. The ISS uses four CMGs. Kepler lost two wheels and had to be reinvented.

NOT THIS Thrusters, which push on expelled gas and use fuel. Wheels only trade angular momentum internally.

CONNECTS Momentum dumping, disturbance torques, attitude sensors.

Homosphere / heterosphere

5

MEANING Below ~100 km the air is mixed to a uniform composition (homosphere). Above, gases separate by molecular weight (heterosphere).

CONCEPT Weather stirs the lower air so thoroughly that its recipe is the same everywhere. Above about 100 km the stirring stops and each gas settles at the height its weight allows: heavy low, light high. That boundary explains why the gas list changes with altitude.

NUMBERS Below 100 km: N_2 78%, O_2 21% everywhere. 200–600 km: atomic O. 600–1,500: He. Above: H.

NOT THIS The temperature layers (troposphere etc.). Homo/heterosphere is about composition; those are about temperature.

CONNECTS Turbopause, atomic oxygen, geocorona.

WORD BANK · 6/8

Concept-level word bank

Atomic oxygen (AO, O) 5

MEANING Single oxygen atoms produced by UV photodissociation of O_2 above ~100 km; the dominant species ~200–600 km. Highly reactive; erodes polymers at orbital velocity.

CONCEPT Up high there's no ozone to absorb UV, so UV splits O_2 into single atoms, and the air is too thin for them to find partners again. The result is a sea of hungry single atoms exactly where LEO satellites fly, and a satellite plowing through it at 7.5 km/s gets chemically sandblasted.

NUMBERS Dominant gas 200–600 km. Bare Kapton on the ISS ram face erodes tens of micrometers per year; the LDEF experiment showed shredded samples after 5.8 years.

NOT THIS Ozone (O_3, three atoms, stratosphere, protective) and O_2 (two atoms, what you breathe). AO is one atom, high up, destructive.

CONNECTS Ram/wake, erosion yield × fluence, coatings, drag (same thin air).

Ionosphere 5

MEANING ~60–1,000 km, overlapping the upper layers: solar UV and X-rays ionize atoms, leaving free electrons that refract and reflect radio and delay GPS signals.

CONCEPT Sunlight's UV and X-rays knock electrons off atoms in the upper atmosphere, leaving a charged layer. Charged layers bend and reflect radio waves, which is why AM radio reaches far at night, why GPS signals get delayed, and why the layer grows and shrinks with day and solar activity.

NUMBERS ~60–1,000 km. GPS delay error up to tens of meters uncorrected; receivers model it out.

NOT THIS The ozone layer (much lower, chemical not electrical) and the magnetosphere (farther out, Earth's magnetic field).

CONNECTS Space weather indices, radio comms section, GPS accuracy.

Specific impulse (Isp) 6

MEANING An engine's fuel efficiency, in seconds: exhaust speed divided by g_0. Chemical rockets 300–450 s; ion engines 3,000+ s with tiny thrust.

CONCEPT Engines differ in how much push they get per kilogram of propellant. Isp is that efficiency, expressed in seconds for historical reasons: multiply by g_0 (9.81) and you get exhaust speed in m/s, which is what the rocket equation actually uses.

NUMBERS Solid rocket ~250 s. Kerosene 300–340. Hydrogen 450. Ion engine 3,000+ (but thrust like the weight of a sheet of paper).

NOT THIS Thrust. Isp is efficiency (miles per gallon); thrust is power (horsepower). Ion engines are high Isp, low thrust.

CONNECTS Rocket equation; propulsion section of the course.

Delta-v (Δv) 6

Also called: velocity change · 'the fuel bill'

MEANING Total change in speed a maneuver or mission requires (km/s). Converts to propellant mass through the rocket equation.

CONCEPT Where you can go isn't limited by distance or time; it's limited by how much you can change your speed, because every speed change costs propellant and propellant is the heaviest thing you carry. So missions are budgeted in km/s, not kilometers.

NUMBERS LEO to GEO: 3.9 km/s. LEO to Mars orbit: ~5.6. Earth surface to LEO: ~9.4 (including losses).

NOT THIS Speed. Δv is the total of all changes, not how fast you're going. A spacecraft can have a huge speed and tiny Δv left.

CONNECTS Rocket equation converts Δv to propellant mass; Hohmann minimizes Δv.

Concept-level word bank

Hohmann transfer

6

Also called: minimum-energy two-impulse transfer

MEANING Half an ellipse tangent to both the starting and target orbit, 180° apart, two prograde burns. Cheapest two-burn transfer between coplanar circular orbits (for radius ratios under ~11.9).

CONCEPT To move between two circular orbits you must change energy twice: once to leave, once to arrive. The cheapest way is an ellipse that just touches both, because both burns are then tangent (no wasted sideways component). It's cheapest, not fastest.

NUMBERS LEO→GEO: 2.42 + 1.47 km/s, 5.3 h. Earth→Mars: ~3.6 from LEO + ~2 to capture, 259 days.

NOT THIS 'Fastest route.' A Hohmann is the slowest sensible route; fast routes cost exponentially more.

CONNECTS Vis-viva for the speeds, time of flight = half the ellipse period, synodic period for when it's possible.

Lambert's problem

6

MEANING Given two position vectors and the time of flight between them, find the orbit that connects them (and so the Δv). The core of trajectory design.

CONCEPT Mission planners don't start with an orbit; they start with 'be at Mars on this date.' Lambert's problem is the tool that turns two places and a deadline into the one orbit that works, and therefore into the fuel bill. It's how every launch date is actually chosen.

NUMBERS Earth on 2026-11-01, Mars on 2027-08-01 → the solver finds the ellipse and reports departure Δv ≈ 3.7 km/s.

NOT THIS Hohmann, which is one special solution (180°, minimum energy). Lambert handles any angle and any time.

CONNECTS Porkchop plots are thousands of Lambert solutions on a grid.

Synodic period

6

Also called: launch-window spacing

MEANING Time between successive identical Earth–target alignments: Earth–Mars ≈ 780 days (26 months).

CONCEPT Two planets circling the Sun at different rates line up the same way only periodically. That repeat time, the synodic period, is why launch windows exist and why a missed Mars window costs two years.

NUMBERS Earth–Mars 780 days (26 months). Earth–Venus 584 days. Earth–Jupiter 399 days.

NOT THIS Orbital period. Mars's year is 687 days; the Earth–Mars lineup repeats every 780 days.

CONNECTS Phase angle: the lineup needed; launch window: the dates around it.

Sphere of influence (SOI)

6

MEANING The region around a planet where its gravity dominates the Sun's for trajectory purposes. Earth's ≈ 925,000 km.

CONCEPT The Sun and a planet both pull on you everywhere, which would make trajectories impossible to compute by hand. The SOI is the practical boundary where you switch from 'planet rules' to 'Sun rules,' so each leg becomes a simple two-body problem.

NUMBERS Earth's SOI ≈ 925,000 km (2.4× the Moon's distance). Jupiter's ≈ 48 million km.

NOT THIS The planet's gravity doesn't stop at the SOI; it's a bookkeeping line, not a physical wall.

CONNECTS Patched conics stitch the legs together at SOI crossings.

WORD BANK · 8/8

Concept-level word bank

Characteristic energy (C3)

6

Also called: launch energy

MEANING C3 = v_∞^2 at Earth departure (km^2/s^2). C3 = 0 is bare escape; Mars needs ~10–15.

CONCEPT Launch providers need one number for 'how hard must I throw this.' C3 is v_∞ squared at Earth departure: zero means barely escaping, bigger means a faster interplanetary departure, and rocket performance charts are drawn against it.

NUMBERS Mars: ~10–15 km^2/s^2. Jupiter direct: ~80. New Horizons: ~158, the highest ever.

NOT THIS Δv. C3 is the energy the rocket must deliver at Earth; Δv is the spacecraft's own budget afterward.

CONNECTS $v_\infty = \sqrt{C3}$; departure burn from LEO = $\sqrt{(v_{esc}^2 + C3)} - v_{circ}$.

Tsiolkovsky rocket equation

6

Also called: ideal rocket equation

MEANING $\Delta v = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$. Links fuel fraction to Δv; why fast transfers are impossible. Coming in the propulsion section.

CONCEPT A rocket pushes on its own exhaust, so it must carry everything it will ever throw. The result is that propellant need grows exponentially with Δv: each extra km/s multiplies the fuel, not adds to it. This single equation explains staging, why fast transfers are impossible, and why Δv is the budget that matters.

NUMBERS Isp 450 s → 4.4 km/s per e-fold. 3.6 km/s needs 56% propellant; 9 km/s (to orbit) needs 87%; 60 km/s needs 99.9999%.

NOT THIS 'More fuel = proportionally more speed.' No: doubling Δv squares the mass ratio.

CONNECTS Specific impulse sets the exhaust speed; Δv comes from the mission plan.

Hyperbolic excess velocity (v∞)

7

MEANING Your speed relative to a planet when far from it, before capture or after escape. The 'in' and 'out' speed of a flyby, equal in magnitude in the planet's frame.

CONCEPT Near a planet, your speed depends on how deep you are in its gravity well, so it's not a useful single number. Far away, gravity's effect has faded and your speed settles to one value: v_∞. It's the clean 'arrival speed' that every flyby and capture calculation starts from.

NUMBERS Earth→Mars Hohmann: v_∞ at Mars ≈ 2.6 km/s. New Horizons left Earth with v_∞ ≈ 12.6 km/s.

NOT THIS Escape velocity. v_∞ is what's left over after you've escaped; escape velocity is the minimum to get $v_\infty = 0$.

CONNECTS $C3 = v_\infty^2$; capture burn size depends on v_∞.

Gravity assist

7

Also called: flyby · swing-by · slingshot

MEANING Passing close to a moving planet so its gravity bends your path; in the Sun's frame you gain (pass behind) or lose (pass in front) speed, borrowed from the planet's orbital motion.

CONCEPT A planet is a moving, massive object. Pass near it and gravity swings you around; because the planet is moving, that swing hands you some of its orbital motion in the Sun's frame. It's the only way to reach the outer planets without impossible amounts of fuel.

NUMBERS Voyager 2 gained ~10 km/s at Jupiter. New Horizons shaved ~3 years off its Pluto trip at Jupiter.

NOT THIS 'Free energy.' The planet pays, by an unmeasurable amount. Also: in the planet's own frame your speed doesn't change at all.

CONNECTS Turning angle formula, hyperbolic excess velocity, momentum conservation.